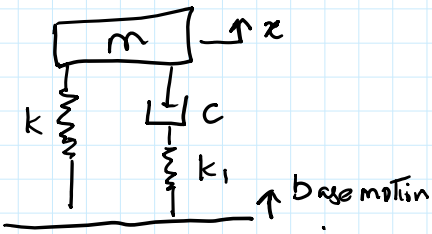
HW2

Saturday, February 14, 2026 3:45 PM



$$m\ddot{x} = -k(x - x_b) - c(\dot{x} - \dot{x}_b) - k_1(-x_b)$$

$$\therefore\ m\ddot{x} + c\dot{x} + (k)x = c\dot{x}_b + (k + k_1)x_b$$

assume $x_b = X_b e^{i\omega t}$ & $x = X e^{i\omega t}$

$$\therefore\ (-m\omega^2 + ic\omega + k)\,X e^{i\omega t} = (ic\omega + k + k_1)\,X_b e^{i\omega t}$$

Transmissibility $T(\omega) = \dfrac{X}{X_b} = \dfrac{k + k_1 + ic\omega}{k - m\omega^2 + ic\omega}$

using $\omega_n = \sqrt{\dfrac{k}{m}}$, $\xi = \dfrac{c}{2\sqrt{km}}$, $r = \dfrac{\omega}{\omega_n}$, $\gamma = \dfrac{k_1}{k}$

$$T(r) = \frac{1 + \gamma + 2i\xi r}{1 - r^2 + 2i\xi r} \quad -(1)$$

$$|T(\omega)| = \sqrt{\frac{(1+\gamma)^2 + (2\xi r)^2}{(1 - r^2)^2 + (2\xi r)^2}} \quad -(2)$$

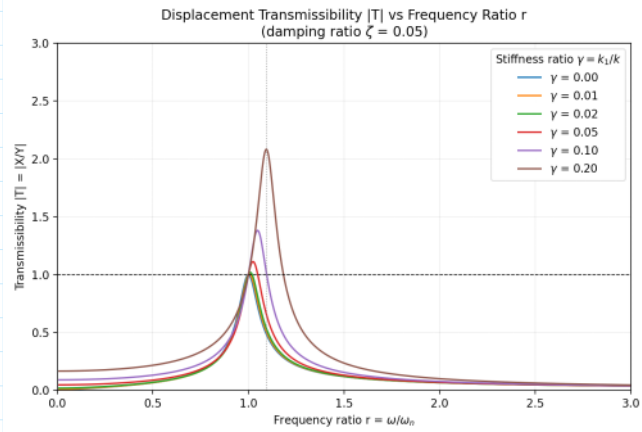


Fig.1 → Transmissibility versus r at various γ

* Case 1 $\gamma = 0$, standard single-spring isolator

$$|T| = \sqrt{\frac{1 + (2\xi r)^2}{(1 - r^2)^2 + (2\xi r)^2}}$$ ≈ isolates very well at high frequencies

* Case 2 small γ ⇝ isolation begins at $r = \sqrt{1+\gamma}$

* case 3 large γ → Transmissibility increases

$\boxed{\text{Hence, case 2 is preferred.}}$

For $\xi = 0.01$ & $\gamma = 0.05$, to find a ξ_{eq} such that

$$\sqrt{\frac{1 + (2\xi_{eq} r)^2}{(1 - r^2)^2 + (2\xi_{eq} r)^2}} \approx \sqrt{\frac{(1.05)^2 + (2 \times 0.01 \times r)^2}{(1 - r^2)^2 + (2 \times 0.01 \times r)^2}}$$

$$\left[1 + (2\xi_{eq} r)^2\right]\left[(1 - r^2)^2 + (2 \times 0.01 \times r)^2\right] \approx \left[1.05^2 + (2 \times 0.01 \times r)^2\right]\left[(1 - r^2)^2 + (2 \times \xi_{eq} \times r)^2\right]$$

At resonance, $r = 1$

$$\left[1 + (2\xi_{eq})^2\right]\left[2 \times 0.01\right]^2 = \left(1.05^2 + (2 \times 0.01)^2\right)(2\xi_{eq})^2$$

$2\xi_{eq} = \alpha$ $\qquad (1 + \alpha^2)(0.02)^2 = 1.1029\,\alpha^2$

$$0.0004 = 1.1025\,\alpha^2$$

$$\alpha = 0.019 \quad \therefore\ \xi_{eq} = 0.0095$$

[for all other areas, a suitable ξ_{eq} cannot be modeled]

$\Rightarrow$ Engine mounts $\rightsquigarrow$ one stiff spring + one soft spring

$\Downarrow$ rubber

$\Downarrow$ hydraulic